CHAPTER

05

Motivation and Emotion

Why did you join university? Why do some students study only before exams while others read every day? Why does your heart pound when you are angry and also when you are afraid? This chapter answers such questions. **Motivation** explains what starts, directs and keeps our behaviour going; **emotion** explains our feelings and how body, behaviour and thought combine to produce them. The words *motive* and *emotion* come from the same Latin root meaning "**to move**".

LESSONS IN THIS CHAPTER

5.1 The Meaning and Types of Motivation

5.2 Theories (Approaches) of Motivation

5.3 Conflict of Motives and Frustration

5.4 Emotion and Theories of Emotion

Freshman A+ · General Psychology (Psyc 1011)

CHAPTER 5 · LESSON 5.1

The Meaning and Types of Motivation



AFTER THIS LESSON YOU CAN

- Define motivation and its three functions
- Distinguish intrinsic and extrinsic motivation
- Distinguish primary and secondary motives
- Recognise each type in real situations

5.1.1 What motivation is

D MOTIVATION

Motivation is the process by which activities are **started (initiated)**, **directed** and **continued (sustained)** so that physical or psychological needs or wants are met. It is what "**moves**" people to act toward a goal. Motivation also affects the **intensity** and **persistence** of behaviour.

HUNGER IN THREE STEPS

You are relaxing in front of the TV and begin to feel hungry. Hunger **starts** the action (you get up), **directs** it (you go to the kitchen, not the bedroom) and **sustains** it (you keep searching, and may even cook, until you have eaten). A motivated person has a goal, increased energy and activity, uses strategies, and **persists**.

5.1.2 Intrinsic and extrinsic motivation

F TWO TYPES OF MOTIVATION

	Intrinsic motivation	Extrinsic motivation
Source	**inside**: the activity itself is enjoyable or satisfying	**outside**: an external reward or to avoid punishment
Examples	reading psychology after the course ends just to deepen your knowledge; running "for fitness and health, not for medals"; a teacher who tutors in her free time because she loves teaching	money for every "A"; a bonus for higher performance; studying only to score an A; running for a prize; helping others to gain acceptance
When the reward is removed	**continues**	tends to **stop**

5.1.3 Primary and secondary motives

- **Primary (biological) motives** come from survival needs and are **unlearned**: hunger, thirst, sex, sleep, avoiding pain.
- **Secondary (acquired, psychological) motives** are **learned** through experience and are **not directly related to survival**: money, social approval, achievement, power, affiliation.

MOTIVATION TRAPS

- "Intrinsic motivation disappears when incentives are removed" is **false**; that describes extrinsic motivation.
- **Money** is a **secondary** motive (learned), even though it buys food.
- "Reading for enjoyment" is **intrinsic**; it cannot be extrinsic.
- The main function of motivation is to **give direction** to behaviour (as well as to start and sustain it), not merely to increase or decrease tension.

Exam Practice

10 QUESTIONS · LESSON 5.1

1 Wolkite University · Final 2015

An internal state or condition that activates and gives direction to our thoughts, feelings and behaviours is known as:

A) incentive
B) primary motive
C) motivation
D) emotion

2 Arsi University · Final 2014

Which of the following is the main function of motivation?

A) Providing feelings
B) Increasing tension
C) Giving direction
D) Decreasing tension

3 Debre Birhan University · Final 2015

Which of the following is NOT true about a motivated person?

A) He or she can increase energy and activity
B) He or she has a goal
C) He or she may or may not persist in the activity
D) He or she uses different strategies to improve performance

4 Wolaita Sodo University · Final 2014

Dana, a student at Wolaita Sodo University, always studies hard because studying gives him personal satisfaction. His desire to read is:

A) intrinsic
B) extrinsic
C) primary
D) secondary

5 Jigjiga University · Final (Set A)

An athlete says, "I run for the sake of fitness and health, not to win medals." This supports:

A) instinct theory
B) incentive theory
C) intrinsic motivation
D) extrinsic motivation

6 Debre Birhan University · Final 2015

Which of the following can NOT be taken as extrinsic motivation?

A) Running for a reward
B) Reading for enjoyment
C) Studying to score an A
D) Helping others to gain acceptance

7 Arba Minch University · Final 2016

Intrinsic motivation will disappear when incentives are removed.

True / False

8 Arba Minch University · Final 2015

A student who reads psychology handouts after finishing the freshman programme to deepen her knowledge shows intrinsic motivation.

True / False

9 Debre Birhan University · Final 2015

Which one of the following is a secondary motive?

A) Food **B)** Sex
C) Water **D)** Money

10 Wolkite University · Final 2015

Motives that are not directly related to biological survival are known as:

A) primary motives **B)** secondary motives
C) instinctive motives **D)** physiological motives

Answers & Explanations

1. C

motivation

Motivation starts, directs and sustains behaviour.

2. C

Giving direction

Motivation directs behaviour toward a goal (it also starts and sustains it).

3. C

He or she may or may not persist in the activity

Motivation includes persistence: a motivated person keeps going toward the goal.

4. A

intrinsic

The activity itself is rewarding to him.

5. C

intrinsic motivation

He is motivated by the internal value of running, not external rewards.

6. B

Reading for enjoyment

Enjoyment of the activity itself is intrinsic motivation.

7. False

Intrinsic motivation does not depend on external incentives; extrinsic motivation does.

8. True

There is no grade or reward left; she reads for the satisfaction of learning.

9. D

Money

Money is learned (acquired); the others are biological.

10. B

secondary motives

Secondary (psychological, acquired) motives are learned and not needed for survival.

CHAPTER 5 · LESSON 5.2

Theories (Approaches) of Motivation

AFTER THIS LESSON YOU CAN

- Explain the instinct, drive-reduction, arousal, incentive, cognitive and humanistic approaches
- Explain needs, drives and homeostasis
- Describe Maslow's hierarchy of needs in order
- Apply each theory to explain a behaviour

5.2.1 Six approaches

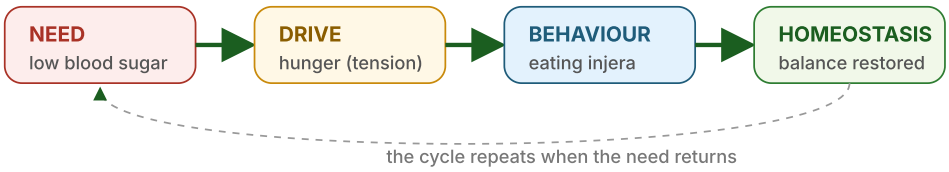
F THE SIX APPROACHES TO MOTIVATION

Approach	Motivation comes from...	Key idea / example	Weakness
Instinct	**inborn, biologically determined** patterns	instinct to reproduce → sexual behaviour; territorial instinct → aggression; birds migrating	only **named** instincts (thousands of them) without explaining them; but showed some behaviour is **hereditary**
Drive-reduction	a **need** creates a **drive** (tension) that pushes us to act and reduce it, restoring **homeostasis**	hungry → eat → hunger reduced	cannot explain why we eat when not hungry or **seek** excitement
Arousal	keeping an **optimal level** of stimulation	too low → seek excitement (sky-diving, horror films); too high → reduce it	
Incentive	**external rewards** that **pull** us	grades, money, affection, attractive food	
Cognitive	**thoughts, beliefs, expectations and goals**	you study as much as you **expect** studying to pay off; distinguishes intrinsic and extrinsic motivation	
Humanistic (Maslow)	a **hierarchy of needs**, from survival to self-actualization	lower needs must be at least partly met before higher ones motivate us	

5.2.2 Needs, drives and homeostasis

D KEY TERMS IN DRIVE-REDUCTION THEORY

- **Need:** a **requirement** for something essential to survival (food, water).
- **Drive:** the **psychological tension and physical arousal** produced by a need; an **internal** state that **pushes** the organism to act.
- **Primary drives** involve survival needs (hunger, thirst); **acquired (secondary) drives** are learned (money, social approval).
- **Homeostasis:** the body's tendency to maintain a **steady internal state**, like a thermostat in a house. A need upsets the balance; behaviour restores it.
  Motivation, according to this theory, begins with **biological needs**.



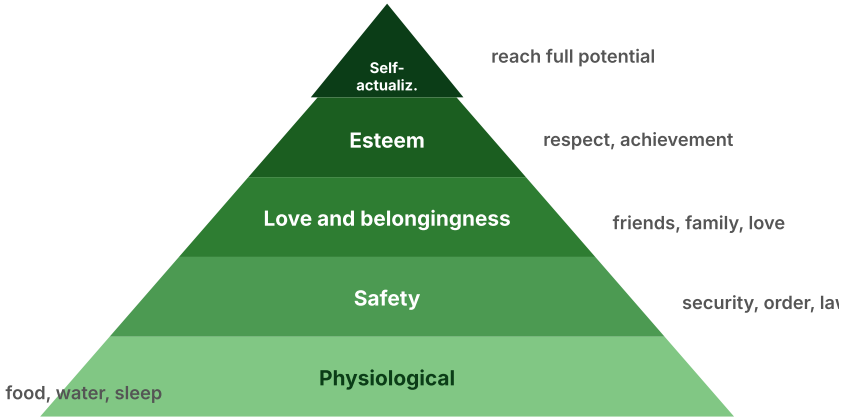
Drive-reduction theory and homeostasis

PUSH AND PULL WORK TOGETHER

Drives **push** from inside; incentives **pull** from outside. At the same moment that hunger pushes you to eat, the smell of fresh bread pulls you toward the bakery. "**When we lack something, we are motivated to get it**" is the drive-reduction view.

5.2.3 Maslow's hierarchy of needs

Abraham **Maslow**, a humanistic psychologist, proposed five levels of needs. Lower needs must be **at least partly satisfied** before higher needs can motivate us.



Maslow's hierarchy of needs (bottom to top)

Level (bottom → top)	Includes	Example
1. **Physiological**	air, food, drink, shelter, clothing, warmth, sex, sleep	a hungry student cannot concentrate on lectures
2. **Safety**	protection, security, order, law, stability, freedom from fear	a person in a conflict zone thinks first of safety
3. **Love and belongingness**	friendship, intimacy, trust, acceptance, family, group membership	a man with a good job and house now looks for a wife
4. **Esteem**	self-esteem (dignity, achievement, mastery, independence) and respect from others (status, prestige)	wanting to be respected as a good student
5. **Self-actualization**	realizing your potential, personal growth, "becoming everything one is capable of becoming"	Maslow's examples: Einstein, Gandhi, (and Mandela)

SELF-ACTUALIZED PEOPLE

Maslow described them as **realistic**, creative, autonomous, accepting of themselves and others, **spiritual** (with peak experiences) and **unconventional**, not selfish or tense. A person with **no job, no home and no friends cannot** become self-actualized, because the lower needs come first.

THEORY TRAPS

- A **drive** is an internal state that **pushes**; an **incentive** is an external stimulus that **pulls**. (A statement that swaps "push" and "pull" is false.)
- Drive-reduction theory focuses on **inner needs**; Maslow's theory describes how some needs take **priority** over others.
- For Maslow, it is the **lower** needs that must be met first, not the higher ones.
- Physiological needs are at the **bottom**; self-actualization is at the **top**.

Exam Practice

12 QUESTIONS · LESSON 5.2

1 Debre Birhan University · Final 2015

_____ are internal states of the organism that push it into action.

A) Needs
B) Deficits
C) Drives
D) Incentives

2 Debre Birhan University · Final 2015

According to drive-reduction theory, motivation begins with:

A) biological needs
B) psychological needs
C) cognitive needs
D) achievement needs

3 Wolkite University · Final 2015

While a drive is an internal state of tension that "pulls" you toward a goal, an incentive is an external stimulus that "pushes" you toward a goal.

True / False

4 Debre Birhan University · Final 2015

In Maslow's hierarchy of needs, the need at the bottom of the ladder is the:

A) safety need
B) self-actualization need
C) love and belonging need
D) physiological need

5 Arsi University · Final 2014

The correct order of Maslow's hierarchy of needs, from lowest to highest, is:

A) physiological, safety, love and belongingness, esteem, self-actualization
B) self-actualization, esteem, love and belongingness, safety, physiological
C) physiological, esteem, safety, love, self-actualization
D) safety, physiological, love, esteem, self-actualization

6 Jigjiga University · Final (Set A)

According to Maslow, a man has a good job and a home and is now looking for a wife. His need relates to:

A) love and belongingness
B) esteem
C) safety
D) self-actualization

7 Debre Birhan University · Final 2015

For Maslow, friends, a sweetheart, children and affectionate relationships in general belong to:

A) physiological needs
B) esteem needs
C) love and belongingness needs
D) self-actualization needs

8 Wolkite University · Final 2015

According to Maslow, a person with no job, no home and no friends can become self-actualized.

True / False

9 Wolaita Sodo University · Final 2014

According to Maslow, self-actualized people are:

A) self-centred, selfish and egocentric
B) realistic, spiritual and unconventional
C) introverted, withdrawn and isolated
D) hard-working, tense and achievement-obsessed

10 Wolaita Sodo University · Final 2014

Which statement about motivation is FALSE?

A) Maslow's hierarchy describes how some needs take priority over others
B) Drive-reduction theory focuses on how inner pushes reduce tension
C) For Maslow, once higher-level needs are met, lower-level needs become important
D) Incentive theory emphasizes external rewards

11 Model question (module discussion)

Your psychology professor says, "Explaining behaviour is easy: when we lack something, we are motivated to get it." Which approach does she follow?

A) Drive-reduction approach
B) Arousal approach
C) Humanistic approach
D) Instinct approach

12 Model question (module)

Bekele, a university student, enjoys bungee jumping and watching horror films because ordinary life feels boring. Which approach best explains his behaviour?

A) Drive-reduction
B) Arousal
C) Instinct
D) Humanistic

Answers & Explanations

1. C

Drives

A drive is the internal tension produced by a need; it pushes behaviour. Incentives pull from outside.

2. A

biological needs

A biological need creates a drive, which motivates behaviour that restores homeostasis.

3. False

It is the reverse: drives push, incentives pull.

4. D

physiological need

Physiological (survival) needs form the base of the hierarchy.

5. A

physiological, safety, love and belongingness, esteem, self-actualization

Survival first, then security, then social needs, then respect, then personal growth.

6. A

love and belongingness

With physiological and safety needs met, he moves on to social needs.

7. C

love and belongingness needs

These are social needs for affection and acceptance.

8. False

Lower needs (physiological, safety, belonging) must be at least partly met before self-actualization.

9. B

realistic, spiritual and unconventional

Maslow described them as realistic, accepting, creative, capable of peak (spiritual) experiences and independent of convention.

10. C

For Maslow, once higher-level needs are met, lower-level needs become important

Maslow said the reverse: lower needs must be met before higher needs become motivating.

11. A

Drive-reduction approach

A lack (need) produces a drive that motivates behaviour to reduce it.

12. B

Arousal

Arousal theory says we seek stimulation when our arousal level is too low; drive-reduction cannot explain seeking excitement.

CHAPTER 5 · LESSON 5.3

Conflict of Motives and Frustration



AFTER THIS LESSON YOU CAN

- Identify the four types of motivational conflict
- Solve scenario questions on conflicts
- Explain why some conflicts cause more stress than others

5.3.1 Four types of conflict

When motives pull us in different directions, we face **conflict**, and failing to reach a goal leads to **frustration**.

THE FOUR MOTIVATIONAL CONFLICTS

Conflict	Situation	Example
Approach–approach	choose between **two desirable** options (only one possible)	going to a movie or a concert; a mother who loves her family but is offered a well-paid job far away, both attractive
Avoidance–avoidance	choose between **two undesirable** options	sell the family home **or** declare bankruptcy; study for a hard exam **or** fail it
Approach–avoidance	**one** option with **both** attractive and unattractive features	a freshman wants to start dating but fears it will consume her study time; loves eating but fears gaining weight; likes an expensive shirt but wants to save money; wants to go abroad to earn money but also wants to stay with family
Multiple approach–avoidance	**two or more** options, **each** with good and bad features	Job A: high salary but long hours and a bad climate; Job B: good benefits and climate but lower pay

SOLVING A CONFLICT QUESTION IN TWO STEPS

"Assefa wants to go abroad to make money and gain experience, but he also needs to stay with his family." Which conflict?

Step 1: **How many options?** Going abroad vs staying. But the question treats going abroad as one goal with a cost (leaving family).

Step 2: **Is each option pleasant, unpleasant or mixed?** The one goal (going abroad) has a good side (money, experience) and a bad side (leaving family) → **approach–avoidance** (the key given by Jigjiga).

Compare: if *both* options are clearly attractive and the question stresses "both situations are attractive" → **approach–approach**. If each of two options has its own pros **and** cons → **multiple approach–avoidance**.

WHICH CONFLICT IS EASIEST?

Approach–approach conflicts are usually the **least stressful**: whatever you choose, you get something good. Avoidance–avoidance and multiple approach–avoidance conflicts are the most difficult.

CONFLICT TRAPS

- In avoidance–avoidance conflict, the two choices are **undesirable**, not "equally important".
- One goal with good **and** bad sides = approach–avoidance; **two or more** such goals = multiple approach–avoidance.

Exam Practice

8 QUESTIONS · LESSON 5.3

1 Arsi University · Final 2014

A person who loves to eat but at the same time fears becoming overweight is most likely experiencing:

A) an approach–avoidance conflict
B) an approach–approach conflict
C) an avoidance–avoidance conflict
D) the fight-or-flight response

2 Debre Birhan University · Final 2015

Mr. X likes the quality of the shirts his friend sells, but knows he should be careful with his money because the shirts are expensive. What type of conflict is he facing?

A) Approach–avoidance
B) Avoidance–avoidance
C) Approach–approach
D) Multiple approach–avoidance

3 Wolaita Sodo University · Final 2014

Mrs. Motuma loves her husband and children very much. She is offered a highly paid job far from her home town, which would force her to leave her family. Both situations are attractive, and she cannot decide. The conflict is:

A) approach–avoidance
B) approach–approach
C) multiple approach–avoidance
D) avoidance–avoidance

4 Jigjiga University · Final (Set A)

Assefa wants to go abroad to make money, gain experience and relax, but at the same time he needs to stay with his family. This conflict is:

A) approach–approach
B) approach–avoidance
C) avoidance–avoidance
D) multiple avoidance

5 Arsi University · Final 2014

In an avoidance–avoidance conflict, a person is forced to choose one of two equally important choices.

True / False

6 Arsi University · Final 2014

Approach–approach conflict causes less frustration than the other conflicts of motives.

True / False

7 Model question (module)

Hirut must choose between two jobs. One offers a high salary with a well-known company but requires long hours and moving to a harsh climate. The other offers promotion opportunities and a better climate but lower pay and an unpredictable schedule. Her conflict is:

A) approach–approach
B) avoidance–avoidance
C) approach–avoidance
D) multiple approach–avoidance

8 Model question (module)

A student must either sit a very difficult make-up exam or accept an F in the course. Which conflict is this?

A) Approach–approach
B) Avoidance–avoidance
C) Approach–avoidance
D) Multiple approach–avoidance

Answers & Explanations

1. A

an approach–avoidance conflict

One goal (eating) is both attractive and unattractive.

2. A

Approach–avoidance

The same shirt attracts him (quality) and repels him (cost).

3. B

approach–approach

The question stresses that both options (family life and the job) are attractive.

4. B

approach–avoidance

Going abroad has attractive features and an unattractive one (leaving his family).

5. False

The two choices are both **undesirable**, not merely equally important.

6. True

Either choice brings something desirable.

7. D

multiple approach–avoidance

There are two options, and each has both positive and negative features.

8. B

Avoidance–avoidance

Both alternatives are undesirable.

CHAPTER 5 · LESSON 5.4

Emotion and Theories of Emotion

AFTER THIS LESSON YOU CAN

- Define emotion and its three components
- Explain the James–Lange, Cannon–Bard and Schachter–Singer theories
- Identify the theory from a statement or diagram

5.4.1 What emotion is

D EMOTION

Emotion is the "**feeling**" aspect of consciousness, characterized by (1) physical **arousal**, (2) **behaviour** that shows the feeling to others, and (3) an **inner awareness** (subjective experience) of the feeling.

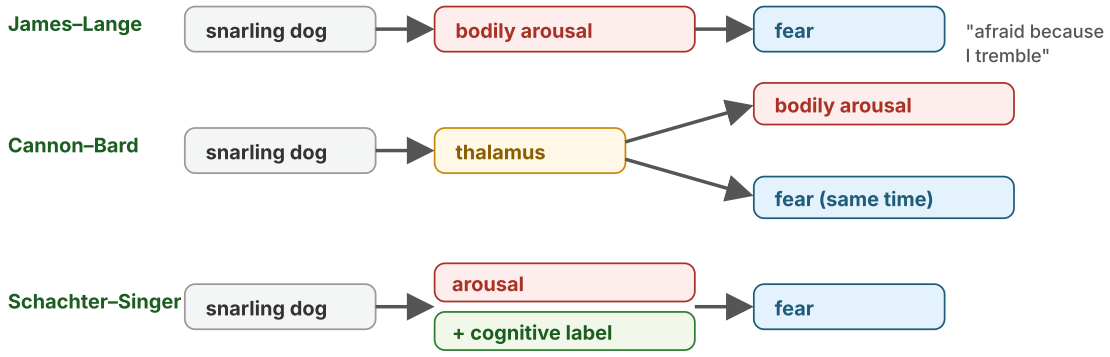
F THE THREE COMPONENTS OF EMOTION

Component	What it is	Examples
Physiological	arousal of the **sympathetic nervous system**	faster heart rate, rapid breathing, dilated pupils, **dry mouth**, shortness of breath, sweating
Behavioural (expressive)	how the emotion shows to others	facial expressions (frown, smile), gestures, body movements, words, actions (fighting, running, hugging, yelling)
Subjective / cognitive	**labelling** the feeling (anger, fear, joy, shame…) using memory and context; partly **learned** and shaped by language and culture	"I feel ashamed"

★ THE BODY ALONE CANNOT TELL EMOTIONS APART

Anger and fear produce **similar** bodily arousal. Without seeing a person's face or knowing the situation, it is easy to mistake fear for anger, which can lead to misunderstanding. Facial expressions differ more, and some expressions are **universal** across cultures, though display rules vary.

5.4.2 Three theories of emotion



The three classic theories of emotion

COMPARING THE THEORIES

Theory	Sequence	In one sentence	Typical statement
James–Lange (William James and Carl Lange)	stimulus → **bodily arousal** → **emotion**	we feel emotion **because of** our physical reaction; emotion depends on physiological responses	"I am afraid **because** my heart is beating fast"; "I am embarrassed because my face is red"
Cannon–Bard (Walter Cannon and Philip Bard)	stimulus → **thalamus** → arousal **and** emotion **at the same time**	bodily changes are too similar to tell emotions apart, so the brain produces the feeling and the arousal **simultaneously**	"I am afraid **and** running **and** aroused!"
Schachter–Singer (two-factor, cognitive arousal)	stimulus → arousal **+ cognitive label** (from cues in the situation) → emotion	emotion needs **both** physical arousal and a **cognitive interpretation** of it	"I am aroused in front of a scary dog, so I **must be afraid**"

ONE RACING HEART, THREE READINGS

Kalkidan's heart races as she waits outside the exam hall.

- **James–Lange:** she notices her pounding heart and sweaty hands, and *therefore* feels anxious.
- **Cannon–Bard:** the pounding heart and the feeling of anxiety arise **together**.
- **Schachter–Singer:** she is aroused, looks around, sees the exam papers and nervous classmates, and **labels** the arousal "anxiety". (If the same arousal happened while waiting to meet a famous athlete, she might label it "excitement".)

EMOTION TRAPS

- **Dry mouth** and **shortness of breath** are the **physiological** component, not the behavioural one.
- "Emotion follows the bodily reaction" = **James–Lange**. "At the same time" = **Cannon–Bard**. "Arousal plus label" = **Schachter–Singer**.
- William James, the founder of **functionalism**, is the same James of the James–Lange theory.

Exam Practice

8 QUESTIONS · LESSON 5.4

1 Debre Birhan University · Final 2015

The theory of emotion that claims emotion happens immediately after physiological arousal is:

A) Cannon–Bard
B) James–Lange
C) Lazarus
D) Schachter–Singer

2 Jigjiga University · Final (Set A)

The statement "I am scared because my heart is beating fast" best illustrates which theory of emotion?

A) Common-sense theory
B) Cannon–Bard
C) Schachter–Singer
D) James–Lange

3 Wolaita Sodo University · Final 2014

Which statement is aligned with the James–Lange theory of emotion?

- **A)** Emotional experiences depend on physiological responses
- **B)** Emotions are objective responses to the environment
- **C)** Emotions result only from cognitive appraisal
- **D)** Emotions and arousal occur independently

4 Debre Birhan University · Final 2015

Shortness of breath and dryness of the mouth are the behavioural component of emotion.

True / False

5 ASTU · Mid 2015

A psychology professor said, "We are psychological beings living in a biological machine, and biological beings living in a psychological machine." What is he trying to convey?

- **A)** The dominant influence of emotions on physiological changes
- **B)** The dominant influence of physiological changes on emotions
- **C)** The independent influence of physiology and emotion on behaviour
- **D)** The interdependent influence of physiology and emotion on behaviour

6 Model question (module)

According to the Cannon–Bard theory:

- **A)** emotion follows bodily arousal
- **B)** bodily arousal follows emotion
- **C)** emotion and bodily arousal occur at the same time
- **D)** emotion requires a cognitive label of arousal

7 Model question (module)

Which theory says that two things must happen before an emotion occurs: physical arousal and a label based on cues from the environment?

- **A)** James–Lange
- **B)** Cannon–Bard
- **C)** Schachter–Singer
- **D)** Drive-reduction

8 Model question (module)

Labelling a feeling as "shame" by recalling similar past experiences and interpreting the situation belongs to which component of emotion?

- **A)** Physiological
- **B)** Behavioural
- **C)** Subjective (cognitive)
- **D)** Sensory

Answers & Explanations

1. **B**

James–Lange

For James and Lange, the bodily reaction comes first and the emotion follows from it.

2. **D**

James–Lange

The emotion (scared) is explained by the bodily change (fast heartbeat).

3. A

Emotional experiences depend on physiological responses

James–Lange: we feel what our body does.

4. False

They are bodily (physiological) changes caused by the sympathetic nervous system.

5. D

The interdependent influence of physiology and emotion on behaviour

Body and mind affect each other; neither works alone.

6. C

emotion and bodily arousal occur at the same time

The thalamus sends signals simultaneously to the cortex (feeling) and the body (arousal).

7. C

Schachter–Singer

Schachter and Singer's two-factor (cognitive arousal) theory.

8. C

Subjective (cognitive)

The subjective component involves interpreting and naming the feeling.

CHAPTER 5 · QUICK REVISION

Motivation and Emotion — at a glance

CHAPTER 5 AT A GLANCE

Topic	Must-know points
Motivation	starts, directs, sustains behaviour; Latin "to move"
Types	intrinsic (inner satisfaction) vs extrinsic (external reward); primary (biological) vs secondary (learned, e.g. money)
Approaches	instinct (inborn); drive-reduction (need → drive → behaviour → homeostasis); arousal (optimal stimulation); incentive (external pull); cognitive (expectations, beliefs); humanistic (Maslow)
Maslow	physiological → safety → love/belonging → esteem → self-actualization
Conflicts	approach–approach (two goods, least stress); avoidance–avoidance (two bads); approach–avoidance (one with good and bad); multiple approach–avoidance
Emotion	feeling aspect: physiology + behaviour + subjective label
Theories	James–Lange (arousal → emotion); Cannon–Bard (simultaneous, thalamus); Schachter–Singer (arousal + cognitive label)

MASLOW FROM THE BOTTOM: "PLEASE STOP LYING, EVERYONE STUDIES"

Physiological, **S**afety, **L**ove, **E**steem, **S**elf-actualization.